

We define the following persuasion percentages to quantify LLM response changes:

$$\begin{aligned}\mathcal{F} &:= 100 \cdot P(R_f = R_r), \\ \mathcal{F}_c &:= 100 \cdot P(R_f = R_r \mid T(R_i) = 1), \\ \mathcal{F}_i &:= 100 \cdot P(R_f = R_r \mid T(R_i) = 0).\end{aligned}\quad (1)$$

Here $\mathcal{F}$ defines the overall percentage that the LLM adopts the challenging response, regardless of correctness, $\mathcal{F}_c$ measures the percentage that the LLM adopts the challenging response given that the initial response was correct, and $\mathcal{F}_i$ measures the percentage that the LLM adopts the challenging response given that the initial response was incorrect.

4 Findings

(H₁) Conversational dynamics amplify persuasion. Table 4 illustrates the persuasion percentages across different models for the Full Rebuttal conversational challenge (FR) and the judge scenarios. Excluding GPT-4o-mini, the results indicate that all models are more likely to adopt the counterargument when it is provided as user input in a second conversational turn compared to when presented in a neutral judge evaluation. Most of the results are statistically significant, rejecting the null hypothesis that persuasion percentages, ($\mathcal{F}$, $\mathcal{F}_c$, $\mathcal{F}_i$; see Eq.1) do not differ with the treatment of FR or Judge (with $p < 0.05$).

Models ↓	Metric →	$\mathcal{F}$ (%)		$\mathcal{F}_c$ (%)		$\mathcal{F}_i$ (%)	
	Challenge →	FR	Judge	FR	Judge	FR	Judge
DeepSeek-V3		36.5	31.7	27.5	22.3	45.6	41.1
GPT-4.1		36.2*	26.5*	23.5*	13.4*	49.0*	39.7*
GPT-4.1-mini		34.4	28.0	20.8*	16.3*	48.1*	39.7*
GPT-4.1-nano		74.6*	66.1*	66.5*	56.1*	80.3	73.6
GPT-4o-mini		37.6*	46.1*	26.8*	35.7*	46.6*	54.5*
Llama-3.3-70B		86.0*	56.5*	80.3*	43.4*	91.6*	69.7*
Llama-4-Maverick		65.1*	40.6*	49.6*	25.7*	80.6*	55.6*
Llama-4-Scout		77.9*	53.4*	66.7*	35.5*	89.1*	71.3*

Table 4: Comparison of persuasion percentages ($\mathcal{F}$, $\mathcal{F}_i$, and $\mathcal{F}_c$; Eq.1) in percentages (three significant figures) for various models across the Full Rebuttal (FR) conversational challenge and the neutral judge experiment. Bold values indicate the higher rate within each comparison pair. An asterisk (*) denotes a statistically significant difference between FR and Judge treatments $\chi^2(1) > 3.841$, $p < 0.05$, under the null hypothesis that ($\mathcal{F}$, $\mathcal{F}_i$, and $\mathcal{F}_c$) do not differ between treatments. All expected cell counts were ≥ 5 . See Appendix §C for full test statistics.

(H₂) Reasoning depth correlates to persuasion. Table 5 reports the persuasion percentage across

different evaluation style rebuttals. The results indicate a clear correlation between the amount of reasoning provided in the challenging rebuttals and the likelihood of the LLM choosing the challenger. For all refutation types and models, all persuasion percentages, ($\mathcal{F}$, $\mathcal{F}_c$, and $\mathcal{F}_i$; Eq.1), increase with more depth of reasoning. This highlights that LLMs are more likely to accept user feedback if reasoning is provided, even when the reasoning is flawed.

(H₃) Style over substance? Dominance of casual assertiveness. Table 6 reports persuasion percentages when LLMs are challenged using various casual challenges. By comparing the average persuasion percentages from casual prompting (Table 6) with those from the evaluation-style Full Rebuttal (FR, average $\mathcal{F} = 56.1\%$, Table 5), we find that casual feedback can be more persuasive, even in the absence of reasoning.

Looking at the average persuasion percentages (Last row of Table 5, Table 6), among the casual styles, the *Sure Rebuttal* (SR) yields the highest overall persuasion percentage ($\mathcal{F}$) of 84.5%. This is considerably higher than the (FR) overall persuasion percentage of 56.1%. The *Are You Sure* (AS) prompt also demonstrate persuasive power similar to those of (FR). The *Divergence Rebuttal* (DR) which provided a concise point of disagreement, has a slightly lower average of $\mathcal{F}$ but is still more persuasive than the Truncated Reasoning. An prominent distinction to note is that DR is the only rebuttal that does not include the proposed answer in its challenge.

These findings suggest that the stylistic nature of the feedback, particularly its casualness and assertiveness, can be a more potent factor in persuading LLMs than the presence or depth of explicit reasoning.

Reasoning quality is a strong predictor of persuasive success. Our findings in H₂ demonstrate that providing a more complete line of reasoning consistently increases a rebuttal’s persuasive power. Prior studies have shown that LLMs prefer longer responses, even if they are of similar qualities (Hu et al., 2025; Saito et al., 2023). These studies show that challenge’s success could be guided by the length of the prompt alone, rather than by the quality of the reasoning.

To investigate this, we focused specifically on the Full Rebuttal (FR) experiment as this is the condition where the rebuttal includes complete

Model ↓	Rebuttal →	Full Rebuttal (FR)			Truncated Rebuttal (TR)			Answer Only Rebuttal (AR)		
	Metric →	$\mathcal{F}$ (%)	$\mathcal{F}_c$ (%)	$\mathcal{F}_i$ (%)	$\mathcal{F}$ (%)	$\mathcal{F}_c$ (%)	$\mathcal{F}_i$ (%)	$\mathcal{F}$ (%)	$\mathcal{F}_c$ (%)	$\mathcal{F}_i$ (%)
DeepSeek-V3		36.5	27.5	45.5	30.9	22.5	39.2	8.1	3.0	13.2
GPT-4.1		36.2	23.5	49.0	17.4	9.6	25.1	15.9	10.1	21.6
GPT-4.1-Mini		34.4	20.8	48.1	22.7	13.5	31.8	9.1	6.9	11.4
GPT-4.1-Nano		74.6	66.5	80.3	63.9	57.6	68.4	19.4	16.5	21.5
GPT-4o-Mini		37.6	26.8	46.6	17.4	13.9	20.1	4.2	2.8	5.3
Llama-3.3-70B		86.0	80.3	91.6	72.4	62.3	82.6	49.6	34.5	64.7
Llama-4-Maverick		65.1	49.6	80.6	57.1	44.2	70.1	49.0	33.6	64.3
Llama-4-Scout		77.9	66.7	89.1	64.5	50.8	78.3	37.7	24.1	51.3
Average		56.1	45.2	66.4	43.3	34.3	51.9	24.1	16.4	31.7

Table 5: Persuasion percentages $\mathcal{F}$, $\mathcal{F}_c$, $\mathcal{F}_i$ (see Eq.1) by model and different degree of reasoning. For all refutation type and model, $\mathcal{F}_c < \mathcal{F}_i$, indicating that **in all scenarios, models are less likely to choose the counterargument if the original answer is correct**. Persuasion rates consistently follow the pattern FR > TR > AR, suggesting that the **inclusion of more reasoning improves persuasive effectiveness, regardless of the correctness of the reasoning**.

Model ↓	Rebuttal →	Are You Sure (AS)			Divergence Rebuttal (DR)			Sure Rebuttal (SR)		
	Metric →	$\mathcal{F}$ (%)	$\mathcal{F}_c$ (%)	$\mathcal{F}_i$ (%)	$\mathcal{F}$ (%)	$\mathcal{F}_c$ (%)	$\mathcal{F}_i$ (%)	$\mathcal{F}$ (%)	$\mathcal{F}_c$ (%)	$\mathcal{F}_i$ (%)
DeepSeek-V3		43.5	27.0	60.1	50.4	38.5	62.4	83.4	69.5	97.2
GPT-4.1		21.6	10.2	33.1	49.6	35.2	64.0	64.3	46.6	82.1
GPT-4.1-Mini		35.0	19.2	50.8	45.4	29.4	61.4	74.7	59.7	89.7
GPT-4.1-Nano		49.9	40.6	56.7	18.6	14.0	21.5	93.9	88.3	98.1
GPT-4o-Mini		25.3	15.7	33.0	26.3	19.5	32.3	71.0	61.2	79.0
Llama-3.3-70B		93.9	88.9	98.9	68.9	59.8	78.0	97.7	97.5	97.8
Llama-4-Maverick		69.2	54.6	83.8	57.8	44.0	71.6	93.0	86.6	99.5
Llama-4-Scout		91.9	84.0	99.7	71.8	64.2	79.5	98.1	96.6	99.5
Average		53.8	42.5	64.5	48.6	38.1	58.8	84.5	75.7	92.9

Table 6: Persuasion percentages $\mathcal{F}$, $\mathcal{F}_c$, $\mathcal{F}_i$ (see Eq.1) across models and casual prompting styles (AS: Are You Sure, DR: Divergence Rebuttal, SR: Sure Rebuttal). in all cases. $\mathcal{F}_c < \mathcal{F}_i$, indicating that **in all scenarios, models are less likely to choose the counterargument when the original answer is correct**. For GPT-4.1-Nano and Llama models, Are You Sure (AS) have higher persuasion percentage than Divergence Rebuttal (DR), suggesting that different models have different cues for sycophantic behaviors. Furthermore, **SR prompts yield the highest persuasion rates** overall, implying that casual assertiveness may be very effective at persuasion.

line of reasoning. We randomly sampled 319 disagreement pairs from our FR results and used an independent LLM judge (Gemini 2.5 flash) to score the quality of both the original ($S_{original}$) and rebuttal ($S_{rebuttal}$) arguments on a 25 point scale. We then analyzed the quality difference $\Delta S = S_{original} - S_{rebuttal}$, against whether the model was persuaded. The prompt for this experiment can be found at Appendix §E.

The results in Table 7 show a clear correlation. When models were persuaded, the rebuttal’s reasoning was, on average, of higher quality (mean $\Delta S = -0.89$). Conversely, when they were not persuaded, the original reasoning was superior (mean $\Delta S = 2.58$). A two-sample t-test confirms this difference is statistically significant ($t = -4.56, p = -7.44e^{-6}$), demonstrating that acceptance to feedback is not only impacted by the

depth of reasoning, as shown in Table 5, but quite unsurprisingly, to the quality of the reasoning steps.

Persuaded	N	Mean ΔS	Std ΔS
TRUE	187	-0.89	7.07
FALSE	132	2.58	6.43

Table 7: Quality Difference Against Persuasion. Models are more likely to be persuaded by rebuttal of higher quality.

Which rebuttals increase accuracy? While persuasion rate measures influence, it doesn’t tell us if the model’s final answer is more accurate. To measure the net impact on accuracy, we define a **Correction Rate** as $\mathcal{F}_i - \mathcal{F}_c$. This metric represents the percentage of times the model correctly changes its response minus the percentage it incorrectly changes its response. A higher value indi-

Style	Persuasion Rate (% $\mathcal{F}$)	Correction Rate ($\mathcal{F}_i - \mathcal{F}_c$)
Judge	43.6%	24.6%
FR	56.1%	21.1%
TR	43.3%	17.6%
AR	24.1%	15.2%
AS	53.8%	22.0%
DR	48.6%	20.8%
SR	84.5%	17.1%

Table 8: Comparison of Persuasion Percentage ($\mathcal{F}$) vs. Correction Rate ($\mathcal{F}_i - \mathcal{F}_c$) by interaction style. The Judge setting serves as a high-performance baseline. Styles are grouped by Evaluation (FR, TR, AR) and Casual (AS, DR, SR) approaches.

cates a more beneficial interaction.

Our analysis in Table 8 reveals two key findings. First, consistent with H_1 , the **Judge** setting provides the highest net accuracy gain (+24.6%), making it the most reliable method for error correction. Second, among rebuttal styles, providing more reasoning leads to better outcomes, with the correction rate for Full Rebuttal (FR) being higher than for Truncated (TR) and Answer-only (AR) rebuttals.

The most persuasive prompt, Sure Rebuttal, shows one of the worst correction rate with a correction rate of 17.1%. While highly effective in persuasion ($\mathcal{F}$: 84.5%), casual assertiveness appears to induce sycophancy indiscriminately, leading to both correct and incorrect changes. For users aiming to correct an LLM, our results suggest the best approach is to reframe the interaction as an evaluation task in a new session.

Overall Trends. Two patterns stand out. First, Llama family consistently demonstrate a high persuasion percentages, with Llama 3.3 70B exhibiting $\mathcal{F} = 93.9\%$ with *Are you Sure* (AS) prompt. This indicates a more pronounced sycophantic tendency in these models. Another consistent observation is that $\mathcal{F}_c < \mathcal{F}_i$ across all conditions. This suggests that LLMs are less likely to revise correct initial answers than incorrect ones.

Persuasion Aggregated by MCQ Datasets. Tables 5 and 6 aggregate persuasion percentage by LLMs and refutation type. Persuasion percentages aggregated by Multiple Choice Question (MCQ) datasets can be found in Table 11. This is to verify whether our results were driven by a particular dataset. While generally consistent, CommonsenseQA exhibits the greatest persuasion percent-

ages in all categories ($\mathcal{F}, \mathcal{F}_c, \mathcal{F}_i$; Eq.1) whereas MMLU shows the lowest persuasion percentages. Interestingly, MMLU also achieves the highest Correction Rate while CommonsenseQA shows the lowest. A closer examination of sycophancy and the nature of the questions may be a worthwhile direction for future work.

5 Conclusion and Future Directions

Conclusion: This research provides a granular analysis of LLM sycophancy in response to second-turn conversational challenges. We find that LLMs are generally more susceptible to persuasion in multi-turn conversation than in neutral evaluation (LLM as a Judge) settings, that the depth of reasoning in a challenge incrementally affects persuasion, regardless of the correctness, and, critically, that the stylistic nature of feedback, particularly casual assertiveness, can be a highly effective tool for persuasion, sometimes outweighing detailed reasoning. These insights are crucial for designing robust LLM interactions and for users to be aware of the dynamics that can influence AI responses.

Future Directions: A deeper dive into the conversation logs, including sentiment analysis of final responses or analysis of the intermediate reasoning steps when a model decides to accept user rebuttal or stand its ground, would be promising. We already observe a departure from the apologetic tone reported by Laban et al. (2024) in older models. Our logs show that LLM seldom apologize. Instead they warp or discard their original reasoning to match user rebuttal.

Limitations

Despite the clear patterns we observe, several factors constrain the scope and generalizability of our findings. Some of them include

Model Coverage. We evaluated a fixed set of contemporary LLMs (GPT-4, 4.1 variants, DeepSeek, and Llama families). Newer, older or models of fundamentally different architectures may exhibit different sycophantic sensitivities. That said, our experimental pipeline can be directly applied to such future or past models.

Task Domain. Our experiments were conducted on multiple-choice questions, which offer a clear right or wrong labels. Open-ended tasks such as short answer generation, essay writing, and dialogue might trigger different sycophantic behaviors.